

Chromosome 3q29 Microduplication Syndrome — Comprehensive Disease Characterization

MONDO: MONDO:0012761 · **Category:** Chromosomal (genomic copy-number) disorder **Report basis:** Aggregated primary literature (case series, cohorts, reviews). No individual EHR data used. **Date:** 2026-09-03

Evidence note. This is a rare disorder described almost entirely through case reports and small cohorts. The largest single cohort to date is 46 patients (

[P 38421086](#)

). Frequencies below are indicative, not population-validated, and most mechanistic claims are inferred from gene content and from the better-studied reciprocal **3q29 deletion**. Statements are flagged where they derive from the deletion rather than the duplication.

1. Disease Information

Overview. Chromosome 3q29 microduplication syndrome is a rare genomic disorder caused by a recurrent copy-number **gain** of ~1.6 Mb at chromosome band 3q29 (hg19 chr3:~195,700,000–197,350,000). It is the reciprocal of the better-characterized 3q29 microdeletion syndrome. The clinical picture is a **variable, generally mild neurodevelopmental disorder**: developmental/speech delay, learning disability or mild–moderate intellectual disability, and a range of accessory features (dysmorphism, ocular and cardiac anomalies, micro- or macrocephaly, obesity). Many carriers are only mildly affected and the duplication is often inherited from a similarly mild parent, so it behaves partly as a **susceptibility factor** rather than a fully penetrant syndrome

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Key identifiers. - **MONDO:** MONDO:0012761 - **OMIM:** 611936 (Chromosome 3q29 microduplication syndrome) - **Orphanet:** ORPHA:284169 (3q29 microduplication syndrome) - **ICD-11:** LD44 (chromosomal duplication category); **ICD-10:** Q92.3 (partial trisomy) - **Reciprocal disorder:** 3q29 microdeletion syndrome — OMIM 609425, Orphanet ORPHA:66634

Synonyms / alternative names: 3q29 duplication syndrome; 3q29 microduplication; trisomy 3q29; dup(3)(q29); 3q29 microduplication syndrome (reciprocal to 3q29 deletion).

Information source type: Disease-level aggregated resources plus published individual case reports; not derived from a single EHR system. A patient registry exists for the reciprocal deletion (3q29deletion.org,

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2. Etiology

Primary cause — genetic (structural). A recurrent interstitial **duplication of ~1.6 Mb at 3q29** containing ~21 protein-coding genes

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Reported duplications range from ~448 kb to ~2.3 Mb, classically spanning **TFRC → BDH1**

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Origin is **non-allelic homologous recombination (NAHR)** between flanking segmental duplications (low-copy repeats): in 89% (16/18) of probands breakpoints fell within paralogous 20-kbp segments inside the 3q29 SDs

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Genetic risk factors. - The recurrent CNV itself is the causal lesion. - **Second-hit CNVs / additional variants** act as risk/modifier factors and were frequent in cohorts (

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additional contributory genetic findings were present in 11/46 patients (

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- Local **segmental-duplication architecture and haplotype diversity** at 3q29 modulate NAHR risk

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Environmental risk factors: None established. No toxin, infectious, dietary, or occupational exposure is implicated in causing the CNV. Parental age effects are not established for this locus.

Protective factors: None specifically identified. Reduced penetrance implies protective genetic-background/modifier effects exist but are uncharacterized.

Gene–environment interactions: No specific GxE interaction documented. Variable expressivity is currently best explained by **genetic modifiers / second-hit variants** (oligogenic model), not by environment

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3. Phenotypes

Phenotype frequencies are approximate (small cohorts, ascertainment bias toward NDD). Duplication features are consistently **less frequent and milder** than in the reciprocal deletion (

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Phenotype	Type	Frequency (indicative)	Onset	Suggested HPO
Speech/language delay	developmental	Common (most frequent)	Childhood	HP:0000750
Global developmental delay	developmental	Common	Infancy/ childhood	HP:0001263
Learning disability	cognitive	High (isolated dup)	Childhood	HP:0001328
Intellectual disability (mild–moderate)	cognitive	Lower than in deletion	Childhood	HP:0001249
Microcephaly	physical	~50% of reported patients	Congenital/ childhood	HP:0000252
Macrocephaly	physical	Subset	Childhood	HP:0000256
Facial dysmorphism	physical sign	Common	Congenital	HP:0001999 / HP:0000271
Ocular abnormalities	physical	Frequent	Congenital/ childhood	HP:0000478
Congenital heart defect	physical	Frequent	Congenital	HP:0001627
Epilepsy/seizures	neurological	Subset	Childhood	HP:0001250
Structural brain anomaly (e.g., gray-matter heterotopia, cortical atrophy)	imaging	Subset	Variable	HP:0002011
Generalized obesity / overweight	metabolic	Subset (mirror trait)	Childhood	HP:0001513
Cleft palate	physical	Uncommon	Congenital	HP:0000175
Musculoskeletal anomalies	physical	Subset	Childhood	HP:0011842
Dental anomalies	physical	Subset	Childhood	HP:0000164

Phenotype	Type	Frequency (indicative)	Onset	Suggested HPO
Hypotonia	neurological	Subset	Infancy	HP:0001252
ASD / autistic features	behavioral	Subset	Childhood	HP:0000717
Recurrent infections	immunologic	Reported (case)	Variable	HP:0002719

- **Phenotype characteristics:** onset predominantly **congenital-to-childhood**; a notable **late-onset (age 34) presentation** with progressive cortical atrophy and recurrent mucosal infections has been reported

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Severity is **variable/mild**; course generally **stable** (developmental), though brain-atrophy and epilepsy subsets may progress.

- **Quality-of-life impact:** driven mainly by cognitive/learning and speech impairment affecting education and daily functioning; cardiac/ocular anomalies may need intervention. No syndrome-specific EQ-5D/SF-36 data published for the duplication.

Core spectrum quote: *"clinical features that include intellectual disability, language delay, epilepsy, structural brain anomalies, micro/macrocephaly, generalized obesity, ocular abnormalities, distinctive facial features, cleft palate, and musculoskeletal anomalies"*

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Psychiatric distinction (important). Large schizophrenia case-control CNV studies associate the **3q29 deletion** — not the duplication — with schizophrenia

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The **duplication** is instead a recurrent susceptibility locus for **autism spectrum disorder and developmental delay**

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with schizophrenia and generalized anxiety being phenotypes unique to deletion carriers (P 39739615).

Counseling should reflect this dosage-direction asymmetry.

4. Genetic / Molecular Information

Causal lesion: recurrent 3q29 duplication (dosage gain), not a point mutation. **Variant class:** structural — copy-number gain (tandem/interstitial duplication). **ACMG/AMP CNV classification:** recurrent 1.6-Mb 3q29 duplication is generally **VUS to likely pathogenic / pathogenic with reduced penetrance** depending on inheritance and second hits; many are inherited from unaffected/mildly affected parents (P 38421086).

Genes in the recurrent interval (~21 protein-coding), key candidates: - **DLG1** (HGNC:2900; SAP97) — synaptic MAGUK scaffolding protein; strongest candidate (P 29501613).

- **BDH1** (HGNC:1027) — mitochondrial 3-hydroxybutyrate dehydrogenase (ketone-body/energy metabolism); smallest single-gene critical region defined (P 24838842).

- **PAK2** (HGNC:8591) — p21-activated kinase; cytoskeletal/synaptic signaling (P 39739615).

- **FBXO45** (HGNC:29148) — synaptic ubiquitin-ligase adaptor, neuronal development (P 29501613).

- **PAK2** (HGNC:8591) — p21-activated kinase; cytoskeletal/synaptic signaling (P 24838842).

Full verified gene inventory (Ensembl GRCh37, chr3:195,700,000–197,350,000; 22 protein-coding genes, this analysis): BDH1, CEP19, DLG1, FBXO45, MFI2/MELTF, NCBP2, NRROS/LRRC33, PAK2, PCYT1A, PIGX, PIGZ, RNF168, SENP5, SLC51A/OSTA, SMC01, TCTEX1D2, TFRC,

TM4SF19, UBXN7, WDR53, ZDHHC19 (+1 putative transcript). This matches the literature "~21 genes"

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Contiguous-gene / dosage-pleiotropy links (candidate): beyond the neurodevelopmental core (DLG1, PAK2, FBXO45, BDH1), the interval carries genes with independent Mendelian disease relevance that plausibly contribute to the pleiotropic phenotype: **CEP19** (recessive morbid obesity / ciliopathy → obesity), **PCYT1A** (retinal dystrophy with spondylometaphyseal dysplasia → ocular anomalies), **RNF168** (RIDDLE-syndrome DNA-damage response → immune/radiosensitivity), **TFRC** (transferrin receptor; combined immunodeficiency → recurrent infections). These are hypothesized dosage contributors, not proven for the duplication.

Minimal critical region: progressively refined from DLG1+BDH1 (448 kb;

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to **BDH1 alone** (single-gene duplications associated with ASD, heart defects, biliary tract dysfunction, obesity;

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Functional consequence: presumed **gain-of-dosage (increased gene expression)** of dosage-sensitive synaptic/metabolic genes; not loss-of-function.

Allele frequency: rare; the recurrent CNV is essentially absent/very rare in gnomAD-SV controls and enriched in NDD cohorts (~1:5000 among NDD patients;

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Somatic vs germline: germline; both **de novo and inherited** (most inherited;

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Post-zygotic occurrence documented as a discordant CNV in monozygotic twins with psychosis

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Modifier genes / epigenetics: second-hit CNVs modify severity

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no locus-specific methylation/histone signature (episignature) is established for the duplication. **Chromosomal abnormality:** dup(3)(q29), submicroscopic — karyotype normal, detected by microarray.

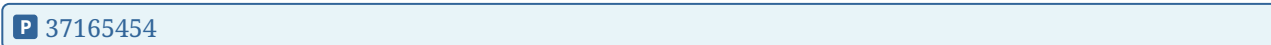
5. Environmental Information

No environmental, lifestyle, or infectious agent is known to cause or trigger 3q29 microduplication syndrome. It is a constitutional genomic disorder. (Obesity within the phenotype may be modulated by diet/lifestyle as in the general population, but this is not disease-specific.) Infectious agents: not applicable.

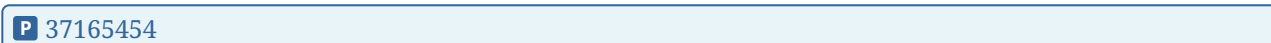
6. Mechanism / Pathophysiology

Causal chain (initiating lesion → clinical manifestation):

1. **Segmental duplications flanking 3q29** provide high sequence identity that **predisposes to misalignment** during meiosis

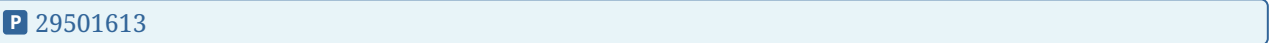
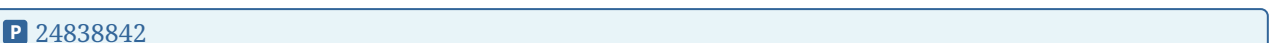
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2. Misalignment **leads to** non-allelic homologous recombination (NAHR) between paralogous ~20-kbp SD segments

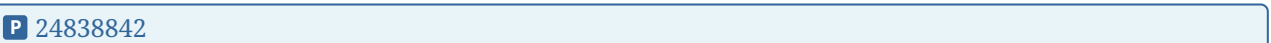
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3. NAHR **results in** a recurrent ~1.6-Mb **duplication** (3 copies) of ~21 genes, transmitted in the germline.

4. Extra gene copies **lead to** increased dosage/expression of dosage-sensitive genes — notably **DLG1, BDH1, PAK2, FBXO45** (inferred;


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5. Elevated **DLG1/PAK2/FBXO45** dosage **is inferred to perturb** synapse assembly, glutamatergic scaffolding and neuronal cytoskeletal signaling (inferred from gene function;

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6. Elevated **BDH1** dosage **is inferred to alter** ketone-body/mitochondrial energy metabolism, potentially affecting neuronal and biliary function (inferred;

(P 39739615).

7. Synaptic + metabolic perturbation **leads to** altered neurodevelopment → developmental/speech delay, learning disability, variable ID, seizures, brain structural anomalies

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8. **Branch:** presence of a **second-hit variant/CNV** shifts the outcome toward more severe/syndromic disease (e.g., cerebral palsy, severe ID); its absence **results in** the common mild phenotype

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9. Reduced penetrance: in a permissive genetic background the dosage gain **may produce** no or minimal clinical effect (inherited-from-unaffected-parent scenario;

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Molecular pathways (candidate): glutamatergic synaptic scaffolding via DLG1/PSD-95 family (MAGUK); PAK2-mediated Rho-GTPase/cytoskeletal signaling; ketone-body metabolism via BDH1. **Cellular processes:** synapse organization, dendritic/spine development, neuronal migration (heterotopia reported,

P 29501613).

Metabolic changes: possible ketone/energy-metabolism shift via BDH1 (interconverts acetoacetate ↔ D-3-hydroxybutyrate; CHEBI:17968 (R)-3-hydroxybutyrate, CHEBI:13705 acetoacetate), and biliary tract dysfunction

(P 39739615);

SLC51A/OSTA in the interval is a bile-acid transporter, offering a candidate link to the reported biliary phenotype. **Immune involvement:** recurrent infections in isolated cases

(P 32874693

— not a core feature.

Model-organism support: direct duplication models are lacking. **Deletion Df/+ mice** show neurodevelopmental/behavioral abnormalities and reduced paraventricular oxytocin neurons, with social deficits rescued by oxytocin (P 35346312) — informative for the locus but reflecting loss, not gain, of dosage.

Suggested ontology terms: GO:0050808 (synapse organization), GO:0007268 (chemical synaptic transmission), GO:0007612/GO:0007611 (learning/memory), GO:0046951 (ketone body biosynthesis); CL:0000540 (neuron), CL:0000679 (glutamatergic neuron).

7. Anatomical Structures Affected

- **Primary organ/system:** central nervous system (UBERON:0001017) — brain (UBERON:0000955); cerebral cortex, lateral ventricles (periventricular heterotopia, P 29501613), cortical atrophy (P 32874693).
- **Secondary organ involvement:** eye (UBERON:0000970) — ocular anomalies; heart (UBERON:0000948) — congenital heart defects; craniofacial skeleton — dysmorphism, cleft palate (UBERON:0001716 secondary palate); musculoskeletal system; teeth; biliary tract (case-level, P 39739615); adipose/metabolic (obesity).
- **Body systems:** nervous, cardiovascular, ocular/visual, musculoskeletal, craniofacial, endocrine/metabolic.
- **Cell/tissue level:** neurons and synapses (nervous tissue) are the principal targets.
- **Subcellular:** postsynaptic density/synapse (GO:0014069), mitochondria (GO:0005739, via BDH1), nucleus.
- **Lateralization:** generally bilateral/symmetric; the reported heterotopia was unilateral (right lateral ventricle, P 29501613).

8. Temporal Development

- **Onset:** congenital to early childhood; developmental delay usually recognized in infancy/toddlerhood. A **late-onset adult** presentation (34 y) is documented

([P 32874693](#)).

Onset pattern: **insidious/chronic** (developmental).

- **Progression:** developmental features are generally **stable** over time; most carriers are non-progressive. Subsets show **progressive** features (cortical atrophy, epilepsy)

([P 32874693](#)).

Duration: **lifelong**.

- **Course pattern:** static/chronic; not relapsing-remitting.
- **Critical periods:** prenatal/early-childhood neurodevelopment is the key window; several diagnoses were made **prenatally** by microarray

([P 33039685](#)),

enabling early developmental intervention.

9. Inheritance and Population

- **Epidemiology:** rare; estimated ~1:5000 among patients with a neurodevelopmental phenotype

([P 39739615](#)).

General-population prevalence is not firmly established; only ~60+ patients are described in the literature (16 early cases + 11 [PMID 33039685] + 46-patient cohort [PMID 38421086] + others).

- **Inheritance pattern:** autosomal, **dominant with reduced penetrance and variable expressivity**; **most cases inherited** from a mildly affected/unaffected parent, some de novo

([P 38421086](#)).

- **Penetrance:** incomplete/reduced

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Expressivity: highly variable

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- **Anticipation:** not established. **Germline/post-zygotic mosaicism:** post-zygotic origin documented in an MZ-twin pair discordant for psychosis

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Founder effects/consanguinity: not relevant (recurrent NAHR event, not a founder mutation). **Carrier frequency:** not defined.

- **Population demographics:** no strong ethnic predilection; cohorts from Europe (France, Romania, Italy), North America, China. **Sex ratio:** roughly balanced/slight male excess in NDD cohorts (~56–62% male in related 3q29 series,

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Age distribution: predominantly children at ascertainment.

10. Diagnostics

- **First-line test: chromosomal microarray analysis (CMA / aCGH / SNP-array)** — the diagnostic gold standard; detects the submicroscopic ~1.6-Mb gain

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Karyotype is normal (lesion below cytogenetic resolution).

- **Confirmation/segregation:** qPCR and **FISH**; parental testing to assess inheritance

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- **Sequencing:** WES/WGS recommended when presentation is **severe or syndromic**, to detect modifying second-hit variants

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Long-read/optical mapping resolves complex SD architecture and breakpoints (research;

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- **Prenatal:** CMA on chorionic villus/amniocentesis samples — several 3q29 duplications identified prenatally

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- **Adjunct workup by phenotype:** brain MRI (heterotopia, atrophy;

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echocardiography (CHD), ophthalmologic exam, developmental/cognitive assessment.

- **Biomarkers:** no specific molecular biomarker; diagnosis is genomic.
- **Differential diagnosis:** other recurrent NDD-associated CNVs (16p11.2, 22q11.2, 1q21.1), 3q29 **deletion** syndrome, and overlapping trisomy 3q / adjacent CNVs (e.g., 1q43q44 overlap,

P 30263904
);

OAVS-associated proximal 3q29 dup,

P 25735547
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Co-occurring monogenic disease can confound (e.g., AMeD/ADH5-ALDH2,

P 41039406
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- **Screening:** not part of newborn screening; **cascade testing** of relatives is appropriate given frequent inheritance.

11. Outcome / Prognosis

- **Survival/mortality:** the syndrome is **not intrinsically life-limiting**; no reduced life expectancy documented for isolated duplications. Mortality risk depends on associated anomalies (e.g., severe congenital heart disease).

- **Morbidity/function:** dominated by **learning disability, speech/language impairment and variable ID**; global adaptive-function deficits are well documented in the reciprocal deletion

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and, more mildly, expected here. Most isolated-duplication patients have **mild NDD**

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- **Complications:** epilepsy, structural brain anomalies, cardiac and ocular disease, obesity, musculoskeletal issues; psychiatric comorbidity possible but **less than in the deletion** (schizophrenia/GAD were unique to deletion carriers,

P 39739615).

- **Recovery/course:** developmental features are stable/chronic; early intervention (speech/OT/PT, special education) improves functional outcome.
- **Prognostic factors:** **presence of a second-hit variant** and additional structural anomalies predict a more severe outcome

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isolated duplication predicts milder outcome.

12. Treatment

No disease-specific or curative therapy exists; management is **symptomatic, multidisciplinary and supportive**.

- **Developmental/behavioral:** early intervention — speech therapy, occupational therapy, physical therapy, special education; behavioral support for ASD/ADHD features (NCIT: Rehabilitation Therapy).
- **Neurologic:** standard **antiseizure medication** for epilepsy (NCIT: Anticonvulsant Agent).
- **Psychiatric:** treat comorbid ADHD/anxiety/psychosis per standard guidelines if present (more relevant to the reciprocal deletion).

- **Cardiac:** surgical/interventional correction of congenital heart defects as indicated (e.g., valvuloplasty) (NCIT: Cardiac Surgery); reciprocal-deletion literature includes balloon valvuloplasty for pulmonary stenosis
([P 36305444](#)).
- **Ophthalmologic/orthopedic/dental:** treat specific anomalies (strabismus/refractive correction, scoliosis/pectus management, dental care).
- **Metabolic:** weight/obesity management.
- **Pharmacogenomics / advanced therapeutics (gene, cell, RNA, targeted, immuno):** none established or in trials specific to this CNV. **Oxytocin** rescued social deficits in the deletion mouse model
([P 35346312](#))
— hypothesis-generating, not a duplication therapy.
- **Genetic counseling:** essential given frequent parental inheritance and reduced penetrance.

13. Prevention

- **Primary prevention:** none possible (constitutional germline CNV). **Genetic counseling** for recurrence risk (up to ~50% transmission from a carrier parent, though penetrance is reduced)
([P 38421086](#)).
- **Reproductive options: prenatal CMA** (documented prenatal diagnoses,
[P 33039685](#))
and **preimplantation genetic testing** for known familial CNV.
- **Secondary prevention:** early developmental screening and intervention; surveillance for treatable anomalies (cardiac, ocular, seizures).
- **Tertiary prevention:** multidisciplinary follow-up to limit complications (educational support, epilepsy control, cardiac monitoring).
- **Cascade screening** of at-risk relatives.
- Immunization / public-health / environmental interventions: not applicable to etiology.

14. Other Species / Natural Disease

- **Taxonomy:** *Homo sapiens* (NCBI:txid9606). No naturally occurring 3q29-duplication disease is described in companion animals or wildlife (OMIA: none specific).
- **Orthologous genes:** the human 3q29 genes have conserved mouse orthologs (Dlg1, Bdh1, Pak2, Fbxo45) enabling engineered models. **Evolutionary conservation:** synaptic scaffolding (DLG1) and PAK signaling are deeply conserved across vertebrates.
- **Comparative biology:** the syntenic region has been engineered in mouse (deletion Df/+;

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Zoonotic potential / transmission: not applicable.

15. Model Organisms

- **Available models:** mouse is the principal system. A **3q29 deletion mouse (Df/+)** recapitulates neurodevelopmental/behavioral abnormalities with high construct and face validity and shows reduced hypothalamic oxytocin neurons

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A dedicated **duplication (gain-of-dosage) mouse model is, to the best of current literature, lacking** — a key gap.

- **Model types:** engineered chromosomal-region models (deletion); single-gene transgenic/knock-in models of DLG1, PAK2, BDH1 exist in general neuroscience literature and could serve as candidate-gene overexpression models.
- **Phenotype recapitulation:** deletion Df/+ mice reproduce social/behavioral deficits (face validity) but model the opposite dosage direction to the duplication; caution in interpretation.
- **Applications:** dissecting synaptic and oxytocinergic mechanisms; testing candidate interventions (e.g., oxytocin,

P 35346312

- **Resources:** MGI (mouse orthologs Dlg1, Bdh1, Pak2, Fbxo45); IMPC/IMSR for single-gene alleles.

Key References (PMID)

- 33039685 — Coyan & Dyer 2020: 11 new cases + literature; second-hit modifiers; prenatal cases.
- 38421086 — Massier et al. 2024: cohort of 46 patients; mild inherited NDD; SRO uncertain.
- 39739615 — Kashevarova et al. 2025: review + BDH1 single-gene critical region; dup vs del comparison; 1:5000.
- 29501613 — Tassano et al. 2018: 448-kb DLG1+BDH1 critical region; heterotopia.
- 24838842 — Fernández-Jaén et al. 2014: severe case; PAK2/DLG1/BDH1/FBXO45 candidates.
- 32874693 — Streata et al. 2020: phenotype heterogeneity; late-onset case.
- 37165454 — Yilmaz et al. 2023: NAHR mechanism, segmental duplications, 21 genes.
- 30263904 — Kessi et al. 2018: overlapping CNV case.
- 25735547 — Guida et al. 2015: proximal 3q29 dup with OAVS.
- 39080272 — Ormond et al. 2024: post-zygotic 3q29 dup in MZ twins.
- 41039406 — 2025: 3q29 dup co-occurring with AMeD (ADH5/ALDH2) syndrome.
- 35346312 — Takemoto et al. 2022: 3q29 **deletion** mouse; oxytocin rescue.
- 36305444 — Kaba & Çelik 2022: reciprocal 3q29 **deletion** case (diagnostic/differential context).
- 37691301 — Pollak et al. 2023: 3q29 registry/musculoskeletal (deletion; registry context).
- 24776740 — Szatkiewicz et al. 2014: Swedish schizophrenia CNV study; 3q29 **deletion** associated with SCZ.
- 22130109 — Grozeva et al. 2012: UK WTCCC CNV frequencies; 3q29 **deletion** enriched in SCZ.
- 22900207 — Nomura & Takumi 2012: review of CNV-based psychiatric animal models; 3q29 among ASD/SCZ loci.
- Ensembl GRCh37 REST (computational, this analysis): 22 protein-coding genes in chr3:195.70–197.35 Mb.

Limitations & Future Directions

- Evidence is dominated by case reports/small cohorts; frequencies are indicative and ascertainment-biased toward NDD.

- The smallest region of overlap and the true driver gene(s) remain unresolved (BDH1 vs DLG1 debated).
- No duplication-specific animal model, no epistatue, and no locus-specific therapy exist.
- Priorities: large genotype–phenotype registries for the duplication, dosage-gain mouse/organoid models, functional dissection of DLG1/BDH1 overexpression, and quantification of penetrance/recurrence risk for counseling.

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